

Aditi Singhania

Embedded Systems & IoT Firmware Engineer • 6 Years Experience • MERIQ Index: 90% (Mastered)

■ aditi.singhania835@example.com | ■ +1 (729) 271-2389 | ■ Toronto, Canada

TECHNICAL PROFICIENCIES & CERTIFICATIONS

Core Technologies: PCB Debugging, Python, C/C++, Embedded Linux, I2C/SPI/UART, FreeRTOS

Verified Certification: Embedded Linux Developer Certificate

PROFESSIONAL WORK EXPERIENCE

Senior Embedded Systems & IoT Firmware Engineer — *Tesla* (2023 - Present (3 yrs))

- Spearheaded architectural redesign of core PCB Debugging service, reducing latency by 60%.
- Led a squad of 5 engineers delivering scalable microservices processing 25M requests/day.
- Implemented automated testing & CI/CD pipelines increasing deployment frequency by 52%.

Associate Embedded Systems & IoT Firmware Engineer — *STMicroelectronics* (2020 - 2023)

- Designed and maintained Python data endpoints with 95% SLA uptime.
- Optimized database indexing and caching layers using C/C++ saving \$27k/yr cloud costs.

NOTABLE TECHNICAL PROJECTS

Smart Grid Power Monitoring Device with MQTT & Hardware Cryptographic Accelerator [PCB Debugging, Python, C/C++, Embedded Linux]

Engineered end-to-end production solution leveraging PCB Debugging and Python with full test automation.

Low-Power BLE Sensor Node Firmware for Industrial Telemetry in FreeRTOS [C/C++, Embedded Linux, I2C/SPI/UART, FreeRTOS]

Architected high-resilience system integrating C/C++ with 99.9% uptime.

EDUCATION & CREDENTIALS

B.S. in Cybersecurity and Information Assurance

UCLA — 2016 - 2020 (Score: 3.5 / 4.0)